Exercise 1 – ML Basics

Introduction to Machine Learning

Exercise 1: Types of learning problems

Learning goals

1. Distinguish supervised from unsupervised tasks (and both from problems where nothing is learned from data)
2. Distinguish regression from classification tasks
3. Identify prediction vs. explanation goals

Below you find short descriptions of five data-related problems. For each scenario, answer the following questions and justify your answers briefly:

- Is this a machine learning problem at all? If so, is it supervised or unsupervised?
- If it is supervised: is it a regression or a classification task?
- Is the primary goal learning to predict or learning to explain?
- For supervised scenarios: what is the target variable (including a sensible target space $\mathcal{Y}$), and what could be suitable features?

a) You work at a second-hand car dealer and want to acquire for-sale vehicles at reasonable prices. To support this, you plan to build a model that predicts a car's adequate market price (in EUR) from its properties, such as age and mileage, using records of past sales.

b) A team of epidemiologists has examined several thousand children, recording for each child whether they have been diagnosed with asthma, along with the average particulate-matter concentration at their home, their age, and whether a parent smokes. They want to quantify how strongly particulate matter affects the risk of a child being asthmatic, accounting for the other influences.

c) A music streaming service wants to organize its catalog into groups of similar-sounding songs, based on audio characteristics such as tempo, energy, and spectral properties. There are no predefined categories the songs should be sorted into.

d) Based on records of past loans with known repayment outcomes, a bank wants to decide on new loan applications by estimating each applicant's probability of repaying. Regulation requires the bank to be able to justify to applicants why they were rejected.

e) A smoke detector should sound an alarm as soon as the measured particle concentration in the air exceeds a threshold defined in a technical standard.

Exercise 2: Car price prediction

> Learning goals
>
> 1) Translate a real-world problem into ML concepts
> 2) Use proper mathematical notation for those concepts
> 3) Decompose a learner into hypothesis space, risk, and optimization

In Exercise 1 a), you characterized the used-car pricing problem as a supervised regression task: you work at a second-hand car dealer and want a model that predicts adequate market prices (in EUR) from vehicles' properties. Let us now develop a complete learner for it.

a) How would you set up your data? Name potential features along with their respective data type and state the target variable.

b) Assume now that you have data on vehicles' age (days), mileage (km), and price (EUR). Explicitly define the feature space $\mathcal{X}$ and target space $\mathcal{Y}$.

c) You choose to use a linear model (LM) for this task. The LM models the target as a linear function of the features with Gaussian error term.

   State the hypothesis space for the corresponding model class. For this, assume the parameter vector θ to include the intercept coefficient.

d) Which parameters need to be learned? Define the corresponding parameter space Θ.

e) State the loss function for the i-th observation using $L2$ loss.

f) Now you need to find the best parameters, and hence the best model, via empirical risk minimization. State the corresponding optimization problem formally.

g) Finally, take a step back: the lecture summarizes supervised learning as *Learning = Hypothesis Space + Risk + Optimization*. Match your results from c) through f) to these three components.

Congratulations, you just designed your first machine learning project!

Exercise 3: Comparing models via empirical risk

The whole exercise is only for lecture group B!

Learning goals

1. Compute pointwise losses and empirical risk by hand
2. Understand how the choice of loss function affects model comparison
3. Explain how risk minimization selects the optimal model

You are back at the second-hand car dealer from Exercise 2. To get a feeling for how loss functions work, you study a small data set of $n = 5$ used cars of one popular vehicle model, with age (in years) as the only feature and price (in 1000 EUR) as the target:

Car i	Age $x^{(i)}$	Price $y^{(i)}$
1	1	16
2	2	14
3	4	10
4	6	6
5	8	10

Car 5 deviates from the general pattern: it is a sought-after classic edition whose price rises again with age.

Two of your colleagues propose two different pricing models:

$$f_1(x) = 18 - 2x, \qquad f_2(x) = 16 - x.$$

a) For both models, compute the predictions $f(x^{(i)})$ for all five cars, as well as the pointwise losses under the $L1$ loss $\left|y^{(i)} - f\left(\mathbf{x}^{(i)}\right)\right|$ and the $L2$ loss $\left(y^{(i)} - f\left(\mathbf{x}^{(i)}\right)\right)^2$.

b) Compute the empirical risk $\mathcal{R}_{\text{emp}}(f)$ of both models under both losses. Which model is preferable under $L1$ loss, and which under $L2$ loss?

c) Explain why the two loss functions disagree about the better model. What role does car 5 play?

d) Our two candidate models form a tiny hypothesis space $\mathcal{H} = \{f_1, f_2\}$. Explain in your own words, and state with a formula, how the principle of empirical risk minimization selects the optimal model $\hat{f}$ from a hypothesis space.

Exercise 4: Vector calculus and the least-squares estimator

The whole exercise is only for lecture group A!

Learning goals

1. Understand how vector-valued functions work
2. Perform calculus on vectors and matrices
3. Derive the least-squares estimator via empirical risk minimization

Consider the following function performing matrix-vector multiplication: $f(\mathbf{x}) = \mathbf{A}\mathbf{x}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^{n \times 1}$.

a) What is the dimension of $f(\mathbf{x})$? Explicitly state the calculation for the i-th component of $f(\mathbf{x})$.

b) Now, consider the gradient (derivative generalized to multivariate functions) $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}$ (a.k.a. $\nabla_{\mathbf{x}} f(\mathbf{x})$).

   (i) What is the dimension of $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}$?

   (ii) Compute $\frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}$.

c) Let's put these rules to use and derive the least-squares estimator you invoked in Exercise 2 f). Recall the ERM problem for the linear model with $L2$ loss:

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta) = \arg\min_{\theta \in \Theta} \sum_{i=1}^{n} \left(y^{(i)} - \theta^\top \mathbf{x}^{(i)} \right)^2,$$

   where, as in Exercise 2, the first entry of each $\mathbf{x}^{(i)}$ is set to 1, so that $\theta \in \mathbb{R}^{p+1}$ contains the intercept. Collect the feature vectors as rows of the design matrix $\mathbf{X} = \left(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\right)^\top \in \mathbb{R}^{n \times (p+1)}$ and the labels in $\mathbf{y} = \left(y^{(1)}, \dots, y^{(n)}\right)^\top$.

   (i) Show that the empirical risk can be written as $\mathcal{R}_{\text{emp}}(\theta) = (\mathbf{y} - \mathbf{X}\theta)^\top (\mathbf{y} - \mathbf{X}\theta)$.

   (ii) Compute the gradient $\frac{\mathrm{d}\mathcal{R}_{\text{emp}}(\theta)}{\mathrm{d}\theta}$.

   Hint: Multiply out the risk first. For the quadratic term, you can use the rule $\frac{\mathrm{d}\theta^\top \mathbf{A}\theta}{\mathrm{d}\theta} = \theta^\top \left(\mathbf{A} + \mathbf{A}^\top\right)$ (try to prove it with the product rule).

(iii) Set the gradient to zero to derive the least-squares estimator $\hat{\theta}$. Which assumption about $\mathbf{X}$ is needed for this solution to exist?

Had trouble with this exercise?

- For the upcoming contents, you need to be able to handle **matrix-valued computations** (multiplication, transposition etc.) and also matrix-valued **calculus**.
- For more explanations and exercises, including a useful collection of rules for calculus, we recommend the book "Mathematics for Machine Learning" (https://mml-book.github.io/book/mml-book.pdf).